NM27C512 Dumper / Original read-only architecture

USB / CH340C / ATmega / clock / reset / ISP / UART

Open sheet 2

File: controller.kicad_sch

Address generator and NM27C512 direct socket

Open sheet 3

File: address_target.kicad_sch

NM27C512 READ-ONLY DUMPER
5 V ONLY
NO VPP PROGRAMMING VOLTAGE

5 V ONLY / NO VPP / no PCB layout

Sheet: /
File: NM27C512_Dumper.kicad_sch

Title: NM27C512 Dumper – read-only baseline

Size: A3	Date: 2026-10-06	**Rev: A**
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USB / CH340C / ATmega / clock / reset / ISP / UART

USB input and internal-clock USB UART

J1
USB Micro-B

F1
500mA hold

PWR_FLAG

U4
CH340C

R4
1k

R5
1k

C1
10uF / 10V

C2
100nF

C3
100nF

U1
ATmega328P-AU

Y1
16MHz, CL -18pF

R1
10k

SW1
RESET

C11
22pF

C12
22pF

C13
100nF auto-reset

C4
100nF

C5
100nF

C6
100nF

C10
100nF

CH340C: no 12 MHz crystal. R232 = GND (TTL mode).
Pins 7/8 are variant-dependent: leave unconnected.
Only DTR# (13) is used; other modem pins are NC.

J2
AVR ISP

J2: 1 MISO, 2 +5V, 3 SCK,
4 MOSI, 5 RESET, 6 GND.
Never power J2 and USB from competing sources.

J3
UART (MCU RX/TX)

J3 names are from MCU viewpoint.
1 = GND 2 = RX (PD0) 3 = TX (PD1)
Use 5 V TTL UART; avoid simultaneous UART drivers.

5 V ONLY / NO VPP / no PCB layout

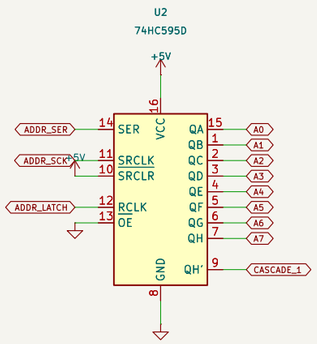
Sheet: /USB / CH340C / ATmega / clock / reset / ISP / UART/
File: controller.kicad_sch

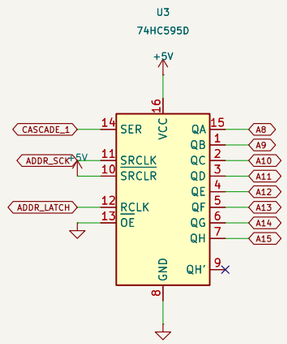
Title: NM27C512 Dumper — read-only baseline

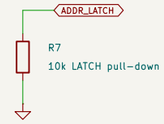
Size: A3	Date: 2026-10-06	**Rev: A**
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Address generator and NM27C512 direct socket

Address / status shift registers: SPI shifts while latch stays LOW



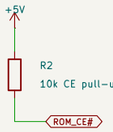


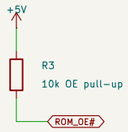


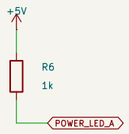


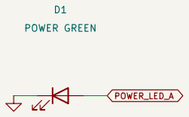


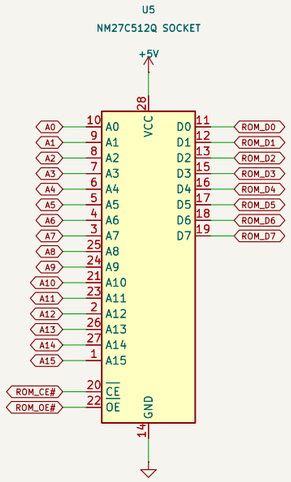
Send U3 (A8–A15) byte first, U2 (A0–A7) byte last; MSB first.
U2 Q0–Q7 = A0–A7; U3 Q0–Q7 = A8–A15.
OE# = GND and SRCLR# = +5V on both 595.
U4 is the CH340C in this read-only project.

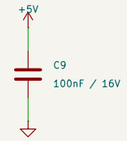












NM27C512 READ-ONLY DUMPER
5 V ONLY
NO VPP PROGRAMMING VOLTAGE
CE# and OE# default HIGH in reset and ISP.

5 V ONLY / NO VPP / no PCB layout

Sheet: /Address generator and NM27C512 direct socket/
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Size: A3	Date: 2026-10-06	**Rev: A**
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